FOXCORE RETAIL (A): DESIGNING A DATABASE

—------- **FINAL PROJECT REPORT** —-------

Designing and Implementing a Relational Database

Stuart School of Business

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INTRODUCTION

Liam Corrigan and Mitchell Fox, two recent graduates, turned a modest idea into a profitable retail business by launching the vibrant Ontario-based startup, Foxcore Retail. The company began as a pop-up shop at music festivals and trade shows, selling novelty products like bubble guns and arctic cooling towels. It quickly gained traction thanks to its energetic presence and appealing product selection. However, as demand grew, so did the complexity of managing the business. Overseeing multiple booths, staff members, and product lines across various events exposed major inefficiencies in Foxcore's manual data tracking systems. The founders soon realized that a more structured and scalable approach was essential to effectively manage their growing operations.

BUSINESS GOALS

Foxcore's primary goals reflect both their immediate operational needs and their long-term vision for growth:

1. **Implement a Structured Sales Tracking System**
   The organization aims to transition from manual, paper-based tracking to a digital database that accurately records inventory movements, sales representative performance, and individual sales transactions.

2. **Improve Operational Efficiency**
   Foxcore aims to minimize errors in commission calculations and event planning, reduce administrative burden, and eliminate data redundancies by centralizing company data and consolidating disparate spreadsheets.

3. **Enable Data-Driven Decision-Making**
   A centralized database would enable the founders to evaluate employee performance, analyze event trends, and identify which products perform best at specific locations—all of which are essential for effective strategic planning.

4. **Support Scalable Growth**
   With a wide range of products and multiple shows operating concurrently, the business needs an infrastructure capable of handling increasing complexity without compromising performance or control.

5. **Enhance Employee Management and Fairness**
   In addition to ensuring fair compensation, accurate sales tracking by employee and shift will help identify high-performing staff and optimize shift assignments for future events.

ENTITIES

1. **Event**
   A scheduled activity held for a specific purpose, such as a trade show, festival, or sports competition.

   Attributes: Event_ID, Venue_ID, Event_Name, Start_Date, End_Date, Description, Event_Type

2. **Venue**
   A location where events take place, equipped with particular features and amenities.

   Attributes: Venue_ID, Venue_Name, Address, Description

3. **Booth**
   A specific space within an event venue designated for showcasing and selling products.

   Attributes: Booth_ID, Booth_Location, Event_ID

4. **Product**
   Items sold at event booths, each having unique features such as type, cost, and selling price.

   Attributes: Product_ID, ProductType, Wholesale_Cost, Minimum_Selling_Price

5. **Salesperson**
   Personnel assigned to sell products at booths, working on scheduled shifts to manage sales.

   Attributes: Salesperson_ID, First_Name, Last_Name, Address, Phone_Number

6. **Sales**
   Records of sales transactions that detail the products sold, the salesperson involved, and the booth and event where the sale occurred.

   Attributes: Sale_ID, Product_ID, Shift_ID, Event_ID, Quantity_Sold, Final_Selling_Price, Date_Time

7. **Shift**
   Details the working hours of a salesperson at a specific booth.

   Attributes: Shift_ID, Salesperson_ID, Booth_ID, Date, Start_Time, End_Time

RELATIONSHIPS

1. **Venue to Event (One-to-Many):**

   One Venue can host none, one or many Events.
   Example: Harborfront Grounds hosts multiple events.

   Each Event must be held only at one Venue.
   Example: Urban Fitness Expo is held at the Metro Convention Centre only.

2. **Event to Booth (One-to-Many):**

   One Event must have at least one Booth**.**
   Example: At Summer Beats Fest, Foxcore has two booths — one selling Bubble Guns and another selling Cooling Towels.

   A Booth can be part of only one Event**.**
   Example: Booth S11 was assigned to the Urban Fitness Expo only.

3. **Shift to Salesperson (One-to-Many):**

   A salesperson must be assigned to at least one Shift.
   Example: Alex works morning and afternoon shifts at different booths during the Urban Fitness Expo.

   Each Shift has to be done by only one salesperson.
   Example: The Saturday afternoon shift at Booth S11 is assigned only to Sam.

4. **Booth to Shift (One-to-Many):**

   A shift has to be assigned to a particular Booth.
   Example: During his Sunday morning shift, Alex worked at Booth S11.ift.

   A Booth must be present at atleast one Shift.
   Example: Booth B3 had separate morning, afternoon, and evening shifts scheduled across the three-day Summer Beats Fest.

5. **Product to Sales (One-to-Many):**

   Each sale has to involve only one product.
   Example: A customer buys a Bubble Gun in a single transaction at the event.

   Each product can be a part of none, one or many sales.
   Example: The Bubble Gun can be sold in over 150 individual transactions at different events.

6. **Event to Sales (One-to-Many):**

   A sale has to be associated with a specific Event.
   Example: The sale of a Bubble Gun is recorded as part of the Summer Beats Fest.

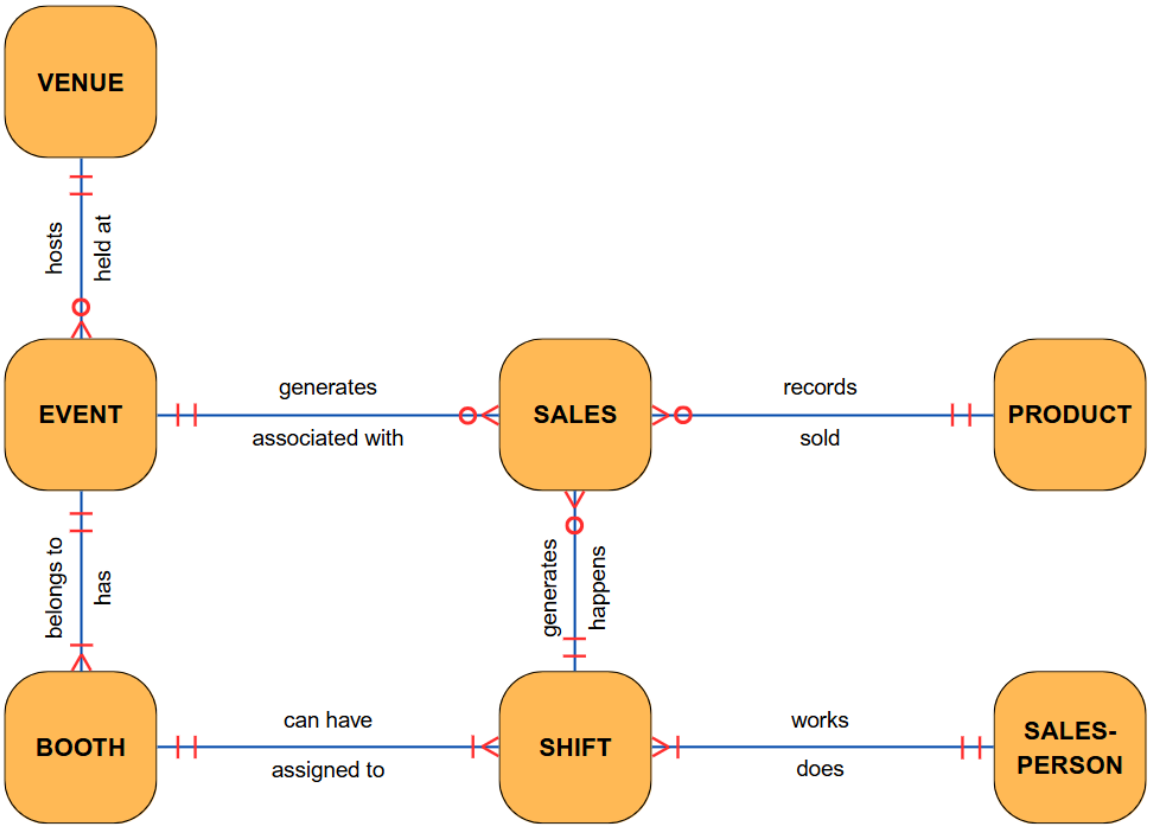
   An Event can have none, one or many Sales.
   Example: Hundreds of sales occur during the three-day Summer Beats Fest.

7. **Shift to Sales (One-to-Many):**

   One Shift can generate none, one or many Sales.
   Example: During Alex's morning shift at Booth S11, he made five different sales of Bubble Guns and Cooling Towels.

   Each Sale must occur during exactly one Shift.
   Example: The sale of a Cooling Towel at 10:15 AM was recorded under Alex's morning shift at Booth S11.

ENTITY RELATIONSHIP DIAGRAM



Assumptions:

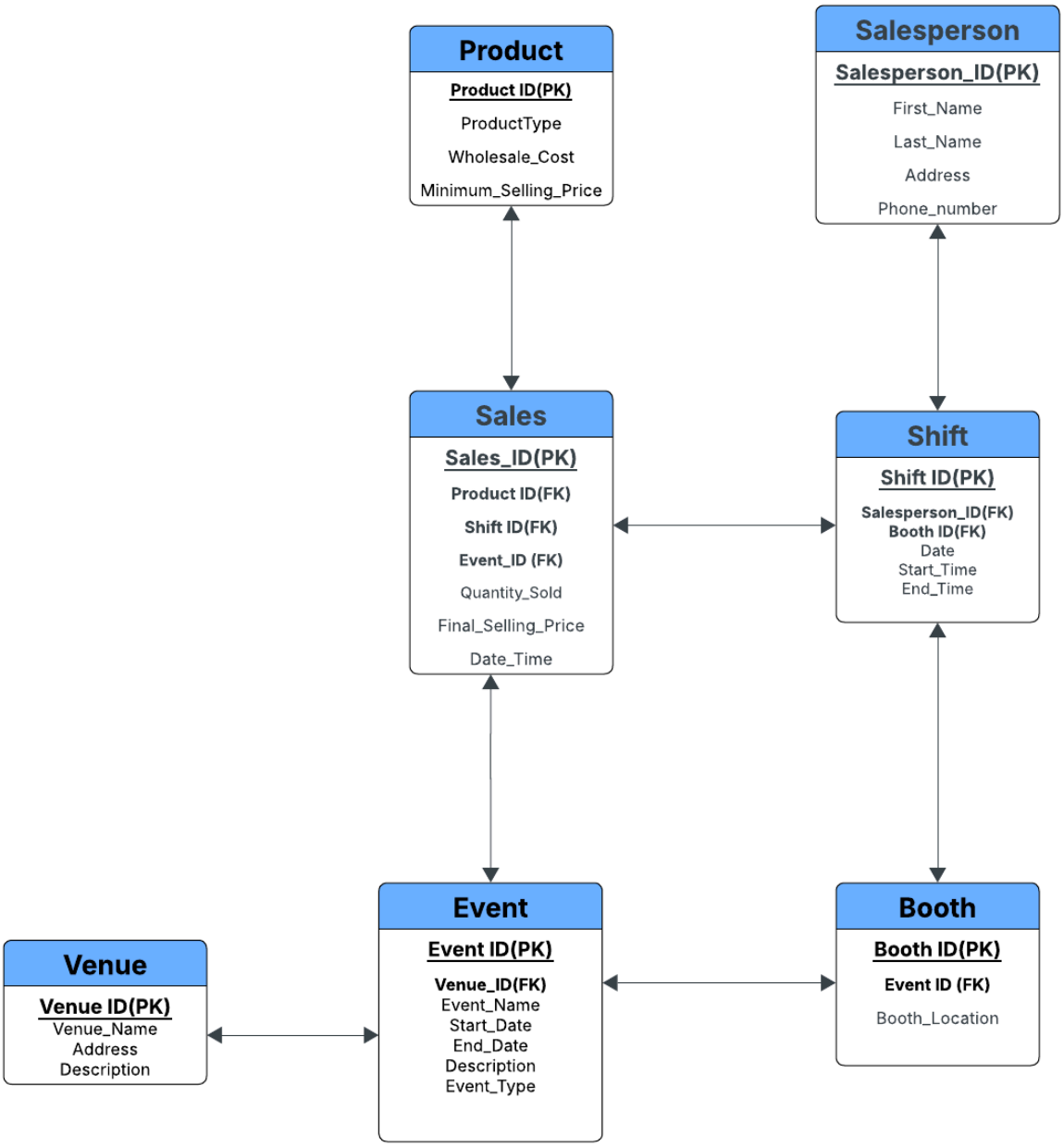
1. **Venue and Event Relationship**
   A venue can host one, multiple or no events at all. .

2. **Product and Sales Relationship**
   A product can have multiple sales or no sales at all.

3. **Sales and Event Relationship**
   An event can record multiple sales or no sales at all.

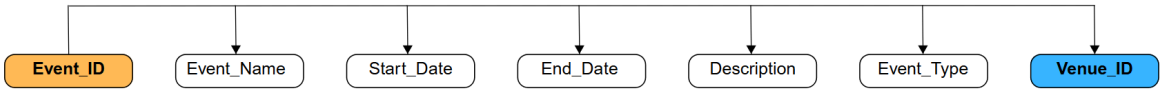
4. **Sales and Shift Relationship**
   There can be none, one, or many sales per shift.

RELATIONAL DATABASE SCHEMA



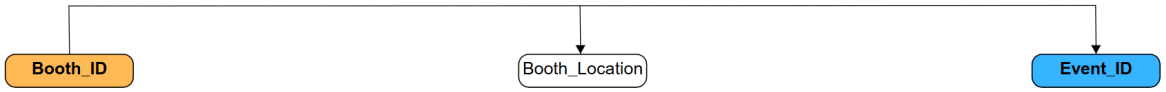
3NF NORMALIZATION DIAGRAM

EVENT

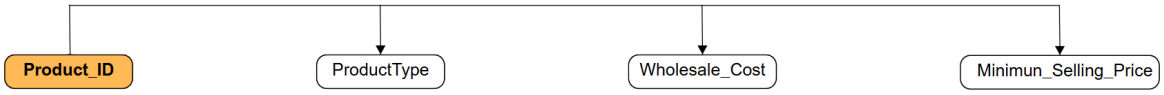


VENUE

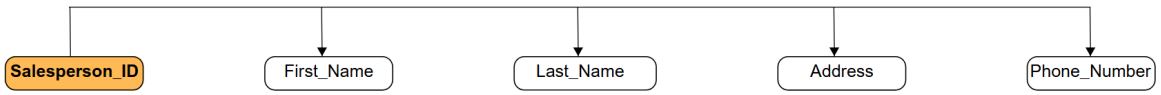
BOOTH



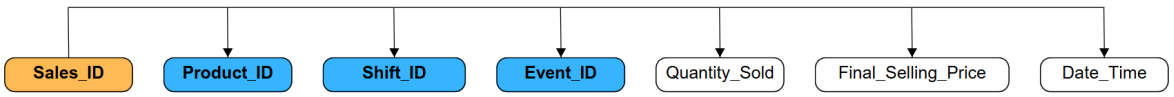
PRODUCT



SALESPERSON



SALES



SHIFT

Each table in the ERD includes a clearly defined **primary key**, such as **Event_ID** in the Event_table or **VenueID** in the Venue table.

All non-key attributes—like **Event_Name**, **Start_Date**, and **Event_Type** in the Event table—are fully functionally dependent on the primary key, fulfilling **Second Normal Form requirements (2NF).**

Additionally, the schema has been designed to eliminate transitive dependencies.
For instance, in the **Sales** table, attributes such as **Quantity_Sold, Final_Selling_Price, and Date_Time** depend only on the primary key **Sale_ID** and not on other non-key attributes.

As a result, the relational schema derived from the ERD adheres to the rules of **Third Normal Form (3NF).**

This level of normalization ensures that the database avoids redundancy and update anomalies, enabling efficient data manipulation and preserving data consistency and integrity across all entities.

DATABASE DICTIONARY

Entity	Attributes	Description	Data type	Nulls	Multi Value
Event	Event_ID	Unique identifier of an event	VARCHAR(9)	No Nulls	No Multi-values
	Venue_ID	Unique identifier of a venue	VARCHAR(9)	No Nulls	No Multi-values
	Event_Name	Name of the event	VARCHAR(50)	No Nulls	No Multi-values
	Start_Date	Start date of the event	DATE	No Nulls	No Multi-values
	End_Date	End date of the event	DATE	No Nulls	No Multi-values
	Description	Description	VARCHAR(150)	No Nulls	No Multi-values
	Event_type	Type of the event	VARCHAR(30)	No Nulls	No Multi-values
Booth	Booth_ID	Unique identifier of a booth	VARCHAR(9)	No Nulls	No Multi-values
	Location	Location of the booth	VARCHAR(25)	No Nulls	No Multi-values
	Event_ID	Unique identifier of an event	VARCHAR(9)	No Nulls	No Multi-values
Venue	Venue_ID	Unique identifier of a venue	VARCHAR(9)	No Nulls	No Multi-values
	Venue_Name	Name of the venue	VARCHAR(50)	No Nulls	No Multi-values
	Address	Location of the event	VARCHAR(150)	No Nulls	No Multi-values

	Description	Description of the venue	VARCHAR(150)	Allows Nulls	No Multi-values
Product	Product_ID	Unique identifier of the product	VARCHAR(9)	No Nulls	No Multi-values
	ProductType	The type of product	VARCHAR(50)	No Nulls	No Multi-values
	Wholesale_ Cost	The wholesale cost of the product	DECIMAL(10,2)	No Nulls	No Multi-values
	Minimum_S elling_Price	The minimum selling price of the product	DECIMAL(10,2)	No Nulls	No Multi-values
Sales-person	Salesperson_ ID	Unique identifier of a salesperson	VARCHAR(9)	No Nulls	No Multi-values
	First_Name	First name of the salesperson	VARCHAR(30)	No Nulls	No Multi-values
	Last_Name	Last name of the salesperson	VARCHAR(30)	No Nulls	No Multi-values
	Address	Address of the salesperson	VARCHAR(150)	Allows Nulls	No Multi-values
	Phone_Num ber	Phone number of the salesperson	INTEGER	Allows Nulls	No Multi-values
Shift	Shift_ID	Unique identifier of a shift	VARCHAR(9)	No Nulls	No Multi-values
	Salesperson_ ID	Unique identifier of a sales	VARCHAR(9)	No Nulls	No Multi-values
	Booth_ID	Unique identifier of a booth	VARCHAR(9)	No Nulls	No Multi-values
	Date	Date of the shift	DATE	No Nulls	No Multi-values
	Start_Time	Start time of the shift	TIME	No Nulls	No Multi-values

	End_Time	End time of the shift	TIME	No Nulls	No Multi-values
Sales	Sale_ID	Unique identifier of a sale	VARCHAR(9)	No Nulls	No Multi-values
	Product_ID	Unique identifier of the product	VARCHAR(9)	No Nulls	No Multi-values
	Shift_ID	Unique identifier of a shift	VARCHAR(9)	No Nulls	No Multi-values
	Event_ID	Unique identifier of an event	VARCHAR(9)	No Nulls	No Multi-values
	Quantity_Sold	Sales quantity of a product	INTEGER	No Nulls	No Multi-values
	Final_Selling_Price	Final Selling Price of the product being sold	DECIMAL(10,2)	No Nulls	No Multi-values
	Date_Time	Date & Time of the sale.	DATETIME	No Nulls	No Multi-values

CREATE STATEMENTS

Event Table

```
CREATE TABLE EVENT (
    Event_ID          VARCHAR(9)          NOT NULL,
    Event_Name        VARCHAR(50)         NOT NULL,
    Start_Date        DATE                 NOT NULL,
    End_Date          DATE                 NOT NULL,
    Description       VARCHAR(150)        NOT NULL,
    Event_Type        VARCHAR(30)         NOT NULL,
    Venue_ID          VARCHAR(9)           NOT NULL,
    CONSTRAINT EVENT_PK PRIMARY KEY (Event_ID),
    CONSTRAINT VENUE_FK FOREIGN KEY (Venue_ID) REFERENCES VENUE (Venue_ID)
);
```

Venue Table

```
CREATE TABLE VENUE (
    Venue_ID        VARCHAR(9)     NOT NULL,
    Venue_Name      VARCHAR(50)    NOT NULL,
    Address         VARCHAR(150)   NOT NULL,
    Description     VARCHAR(150),

    CONSTRAINT PK_Venue PRIMARY KEY (Venue_ID)
);
```

Booth Table

```
CREATE TABLE BOOTH (
    Booth_ID    VARCHAR(9)     NOT NULL,
    Location    VARCHAR(25)    NOT NULL,
    Event_ID    VARCHAR(9)     NOT NULL,

    CONSTRAINT BOOTH_PK PRIMARY KEY (Booth_ID),
    CONSTRAINT EVENT_PK FOREIGN KEY (Event_ID) REFERENCES EVENT (Event_ID)
);
```

Product Table

```
CREATE TABLE PRODUCT (
    Product_ID             VARCHAR(9)     NOT NULL,
    ProductType            VARCHAR(50)    NOT NULL,
    Wholesale_Cost         DECIMAL(10, 2) NOT NULL,
    Minimum_Selling_Price  DECIMAL(10, 2) NOT NULL,

    CONSTRAINT PRODUCT_PK PRIMARY KEY (Product_ID)
);
```

Salesperson Table

```
CREATE TABLE Salesperson (
    Salesperson_ID VARCHAR(9) NOT NULL,
    First_Name VARCHAR(30) NOT NULL,
    Last_Name VARCHAR(30) NOT NULL,
    Address VARCHAR(150),
    Phone_Number BIGINT,

    CONSTRAINT Salesperson_PK PRIMARY KEY (Salesperson_ID)
);
```

Sales Table

```
CREATE TABLE Sales (
    Sale_ID VARCHAR(9) NOT NULL,
    Product_ID VARCHAR(9) NOT NULL,
    Shift_ID VARCHAR(9) NOT NULL,
    Event_ID VARCHAR(9) NOT NULL,
    Quantity_Sold INT NOT NULL,
    Final_Selling_Price DECIMAL(10,2) NOT NULL,
    Date_Time DATETIME NOT NULL,

    CONSTRAINT Sales_PK PRIMARY KEY (Sale_ID),
    FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID),
    FOREIGN KEY (Shift_ID) REFERENCES Shift(Shift_ID),
    FOREIGN KEY (Event_ID) REFERENCES Event(Event_ID)
);
```

Shift Table

```sql
CREATE TABLE Shift (
    Shift_ID VARCHAR(9) NOT NULL,
    Salesperson_ID VARCHAR(9) NOT NULL,
    Booth_ID VARCHAR(9) NOT NULL,
    Date DATE NOT NULL,
    Start_Time TIME NOT NULL,
    End_Time TIME NOT NULL,

    CONSTRAINT Shift_PK PRIMARY KEY (Shift_ID),
    FOREIGN KEY (Salesperson_ID) REFERENCES Salesperson(Salesperson_ID),
    FOREIGN KEY (Booth_ID) REFERENCES Booth(Booth_ID)
);
```

INSERT STATEMENTS

Event

```
INSERT INTO EVENT (Event_ID, Event_Name, Start_Date, End_Date, Description, Event_Type, Venue_ID)
VALUES ('EV1', 'Summer Beats Fest', '2017-07-15', '2017-07-17', 'Live Music and Food Trucks', 'Music Festival', 'VN1');

INSERT INTO EVENT (Event_ID, Event_Name, Start_Date, End_Date, Description, Event_Type, Venue_ID)
VALUES ('EV2', 'Urban Fitness Expo', '2017-08-12', '2017-08-13', 'Fitness Demos and Gear Sales', 'Sporting Event', 'VN2');
```

Event_ID	Event_Name	Start_Date	End_Date	Description	Event_Type	Venue_ID
EV1	Summer Beats Fest	2017-07-10	2017-07-17	Live Music and Food Trucks	Music Festival	VN1
EV2	Urban Fitness Expo	2017-08-12	2017-08-13	Fitness Demos and Gear Sales	Sporting Event	VN2

Venue

```
INSERT INTO VENUE (Venue_ID, Venue_Name, Address, Description)
VALUES ('VN1', 'Lakeside Pavilion', '789 Shoreline Dr, Toronto, ON M4C 1Z2 Canada', 'Scenic Outdoor Venue');

INSERT INTO VENUE (Venue_ID, Venue_Name, Address, Description)
VALUES ('VN2', 'Expo Hall West', '12 King St, Toronto, ON M7B 2K4 Canada', NULL);
```

Venue_ID	Venue_Name	Address	Description
VN1	Lakeside Pavilion	789 Shoreline Dr, Toronto, ON M4C 1Z2 Canada	Scenic Outdoor Venue
VN2	Expo Hall West	12 King St, Toronto, ON M7B 2K4 Canada	NULL

Booth

```
INSERT INTO BOOTH ('Booth_ID', 'Location', 'Event_ID')
VALUES ('BT1', 'L11', 'EV1');

INSERT INTO BOOTH ('Booth_ID', 'Location', 'Event_ID')
VALUES ('BT2', 'L12', 'EV2');
```

Booth_ID	Location	Event_ID
BT1	L11	EV1
BT2	L12	EV2

Product

```
INSERT INTO PRODUCT (Product_ID, ProductType, WholesaleCost, Minimum_Selling_Price)
VALUES ('PT3', 'Glow Stick Set', 2.50, 10.00);

INSERT INTO PRODUCT (Product_ID, ProductType, WholesaleCost, Minimum_Selling_Price)
VALUES ('PT4', 'Cooling Towel', 4.00, 15.00);
```

Product_ID	ProductType	WholesaleCost	Minimum_Selling_Price
PT3	Glow Stick Set	2.50	10.00
PT4	Cooling Towel	4.00	15.00

Salesperson

```
INSERT INTO SALESPERSON (Salesperson_ID, First_Name, Last_Name, Address, Phone_Number)
VALUES ('SP1', 'Alicia', 'Reed', '1020 Carlton St, Toronto, Canada', 4163427888);

INSERT INTO SALESPERSON (Salesperson_ID, First_Name, Last_Name, Address, Phone_Number)
VALUES ('SP2', 'Leo', 'Turner', '389 Queen St W, Toronto, Canada', 4379211164);
```

Salesperson_ID	First_Name	Last_name	Address	Phone_Number
SP1	Alicia	Reed	1020 Carlton St, Toronto, Canada	4163427888
SP2	Leo	Turner	389 Queen St W, Toronto, Canada	43797211164

Sales

```
INSERT INTO SALES (Sales_ID, Product_ID, Shift_ID, Event_ID, Quantity_Sold, Final_Selling_Price, Date_Time)
VALUES ('S1', 'PT3', 'SF1', 'EV1', 12, 10.00, '2017-07-15 11:45:20');

INSERT INTO SALES (Sales_ID, Product_ID, Shift_ID, Event_ID, Quantity_Sold, Final_Selling_Price, Date_Time)
VALUES ('S2', 'PT4', 'SF2', 'EV2', 7, 15.00, '2017-08-12 14:32:10');
```

Sales_ID	Product_ID	Shift_ID	Event_ID	Quantity_Sold	Final_Selling_Price	Date_Time
S1	PT3	SF1	EV1	12	10.00	2017-07-15 11:45:20
S2	PT4	SF2	EV2	7	15.00	2017-08-12 14:32:10

Shift

```sql
INSERT INTO SHIFT (Shift_ID, Salesperson_ID, Booth_ID, Date, Start_Time, End_Time)
VALUES ('SF1', 'SP1', 'BT1', '2017-07-15', '11:00:00', '14:00:00');

INSERT INTO SHIFT (Shift_ID, Salesperson_ID, Booth_ID, Date, Start_Time, End_Time)
VALUES ('SF2', 'SP2', 'BT2', '2017-08-12', '14:30:00', '17:30:00');
```

Shift_ID	Salesperson_ID	Booth_ID	Date	Start_Time	End_Time
SF1	SP1	BT1	2017-07-15	11:00:00	14:00:00
SF2	SP2	BT2	2017-08-12	14:30:00	17:30:00

UPDATE STATEMENTS

Event

```
UPDATE EVENT
SET Description = 'Exciting live music performances and gourmet food trucks'
WHERE Event_ID = 'EV1';
```

Event_ID	Event_Name	Start_Date	End_Date	Description	Event_Type	Venue_ID
EV1	Summer Beats Fest	2017-07-10	2017-07-17	Exciting live music performances and gourmet food trucks	Music Festival	VN1
EV2	Urban Fitness Expo	2017-08-12	2017-08-13	Fitness Demos and Gear Sales	Sporting Event	VN2

Venue

```
UPDATE VENUE
SET Description = 'Beautiful outdoor space near the lakeside'
WHERE Venue_ID = 'VN1';
```

Venue_ID	Venue_Name	Address	Description
VN1	Lakeside Pavilion	789 Shoreline Dr, Toronto, ON M4C 1Z2 Canada	Beautiful outdoor space near the lakeside
VN2	Expo Hall West	12 King St, Toronto, ON M7B 2K4 Canada	NULL

Booth

```
UPDATE BOOTH
SET Location = 'L13'
WHERE Booth_ID = 'BT1';
```

Booth_ID	Location	Event_ID
BT1	L13	EV1
BT2	L12	EV2

Product

```
UPDATE PRODUCT
SET Wholesale_Cost = 3.00
WHERE Product_ID = 'PT3';
```

Product_ID	ProductType	WholesaleCost	Minimum_Selling_Price
PT3	Glow Stick Set	3.00	10.00
PT4	Cooling Towel	4.00	15.00

Salesperson

```
UPDATE SALESPERSON
SET Address = '1050 New Carlton St, Toronto, ON M4C 1Z2, Canada'
WHERE Salesperson_ID = 'SP1';
```

Salesperson_ID	First_Name	Last_name	Address	Phone_Number
SP1	Alicia	Reed	1050 New Carlton St, Toronto, ON M4C 1Z2, Canada	4163427888
SP2	Leo	Turner	389 Queen St W, Toronto, Canada	43797211164

Sales

```
UPDATE SALES
SET Final_Selling_Price = 11.00
WHERE Sale_ID = 'S1';
```

Sales_ID	Product_ID	Shift_ID	Event_ID	Quantity_Sold	Final_Selling_Price	Date_Time
S1	PT3	SF1	EV1	12	11.00	2017-07-15 11:45:20
S2	PT4	SF2	EV2	7	15.00	2017-08-12 14:32:10

Shift

```
UPDATE SHIFT
SET Start_Time = '12:00:00'
WHERE Shift_ID = 'SF1';
```

Shift_ID	Salesperson_ID	Booth_ID	Date	Start_Time	End_Time
SF1	SP1	BT1	2017-07-15	12:00:00	14:00:00
SF2	SP2	BT2	2017-08-12	14:30:00	17:30:00

DELETE STATEMENTS

Event

```
DELETE FROM EVENT
WHERE Event_ID = 'EV2';
```

Event_ID	Event_Name	Start_Date	End_Date	Description	Event_Type	Venue_ID
EV1	Summer Beats Fest	2017-07-10	2017-07-17	Exciting live music performances and gourmet food trucks	Music Festival	VN1

Venue

```
DELETE FROM VENUE
WHERE Venue_ID = 'VN2';
```

Venue_ID	Venue_Name	Address	Description
VN1	Lakeside Pavilion	789 Shoreline Dr, Toronto, ON M4C 1Z2 Canada	Updated Scenic Outdoor Venue

Booth

```
DELETE FROM BOOTH
WHERE Booth_ID = 'BT2';
```

Booth_ID	Location	Event_ID
BT1	L13	EV1

Product

```
DELETE FROM PRODUCT
WHERE Product_ID = 'PT4';
```

Product_ID	ProductType	WholesaleCost	Minimum_Selling_Price
PT3	Glow Stick Set	3.00	10.00

Salesperson

```
DELETE FROM SALESPERSON
WHERE Salesperson_ID = 'SP2';
```

Salesperson_ID	First_Name	Last_name	Address	Phone_Number
SP1	Alicia	Reed	1020 Carlton St, Toronto, Canada	4163427888

Sales

```
DELETE FROM SALES
WHERE Sale_ID = 'S2';
```

Sales_ID	Product_ID	Shift_ID	Event_ID	Quantity_Sold	Final_Selling_Price	Date_Time
S1	PT3	SF1	EV1	12	11.00	2017-07-15 11:45:20

Shift

```
DELETE FROM SHIFT
WHERE Shift_ID = 'SF2';
```

Shift_ID	Salesperson_ID	Booth_ID	Date	Start_Time	End_Time
SF1	SP1	BT1	2017-07-15	12:00:00	14:00:00

SELECT STATEMENTS

Total Revenue and Units Sold by Product at Each Event

Useful to determine which products perform best at different shows.

```
SELECT
    e.Event_Name,
    p.ProductType,
    SUM(s.Quantity_Sold) AS Total_Units_Sold,
    SUM(s.Quantity_Sold * s.Final_Selling_Price) AS Total_Revenue
FROM SALES s
JOIN PRODUCT p ON s.Product_ID = p.Product_ID
JOIN EVENT e ON s.Event_ID = e.Event_ID
GROUP BY e.Event_Name, p.ProductType;
```

Sales and Commission Report per Salesperson

To track performance and calculate commissions.

```
SELECT
    sp.Salesperson_ID,
    sp.First_Name,
    sp.Last_Name,
    SUM(s.Quantity_Sold * s.Final_Selling_Price) AS Total_Sales,
    SUM(s.Quantity_Sold * s.Final_Selling_Price) * 0.10 AS Estimated_Commission
FROM SALES s
JOIN SHIFT sh ON s.Shift_ID = sh.Shift_ID
JOIN SALESPERSON sp ON sh.Salesperson_ID = sp.Salesperson_ID
GROUP BY sp.Salesperson_ID, sp.First_Name, sp.Last_Name;
```

Total Revenue by Event Type

Helps Foxcore decide which event types are most profitable.

```
SELECT
    e.Event_Type,
    SUM(s.Quantity_Sold * s.Final_Selling_Price) AS Total_Revenue
FROM SALES s
JOIN EVENT e ON s.Event_ID = e.Event_ID
GROUP BY e.Event_Type;
```

Sales Per Shift by Salesperson

Supports scheduling, staffing efficiency, and accountability.

```
SELECT
    sh.Shift_ID,
    sp.First_Name,
    sp.Last_Name,
    sh.Start_Time,
    sh.End_Time,
    SUM(s.Quantity_Sold * s.Final_Selling_Price) AS Shift_Sales
FROM SHIFT sh
JOIN SALES s ON sh.Shift_ID = s.Shift_ID
JOIN SALESPERSON sp ON sh.Salesperson_ID = sp.Salesperson_ID
GROUP BY sh.Shift_ID, sp.First_Name, sp.Last_Name, sh.Start_Time, sh.End_Time;
```

Sales Breakdown by Booth and Event

Useful to compare booth placement or layout effectiveness.

```
SELECT
    e.Event_Name,
    b.Booth_ID,
    b.Location,
    SUM(s.Quantity_Sold * s.Final_Selling_Price) AS Booth_Revenue
FROM SALES s
JOIN SHIFT sh ON s.Shift_ID = sh.Shift_ID
JOIN BOOTH b ON sh.Booth_ID = b.Booth_ID
JOIN EVENT e ON s.Event_ID = e.Event_ID
GROUP BY e.Event_Name, b.Booth_ID, b.Location;
```

RECOMMENDATIONS & CONCLUSION

The rapid transformation of Foxcore Retail from a simple festival pop-up to a multi-event retail operation highlights both the entrepreneurial success of its founders and the challenges of scaling a data-driven business. While popular products and a dedicated sales team have helped the company gain significant traction, reliance on manual processes previously led to inefficiencies, fragmented data, and missed strategic opportunities. As operations became increasingly complex, implementing an integrated and reliable database system became essential. Beyond improving day-to-day operations, adopting a structured data management approach provides the analytical foundation necessary for sustained, long-term growth.

With the database system now in place, Foxcore Retail is positioned to overcome its operational challenges and unlock new opportunities for strategic insight. The system streamlines sales tracking, ensures accurate commission calculations, and offers detailed reporting on the performance of events, products, and employees. By consolidating and centralizing data, Foxcore can more effectively manage inventory, identify sales trends, and make informed, data-driven business decisions that fuel continued growth.

It is recommended that Foxcore Retail continue to invest in optimizing its database system. This includes regularly analyzing the collected data to refine operational strategies, forecast demand, and improve employee performance. Ongoing staff training in database usage and best practices will also be essential to maximize the system's benefits. By leveraging the database's full capabilities and fostering a data-driven culture, Foxcore will strengthen its adaptability, competitiveness, and ability to achieve both short-term efficiencies and long-term business success.